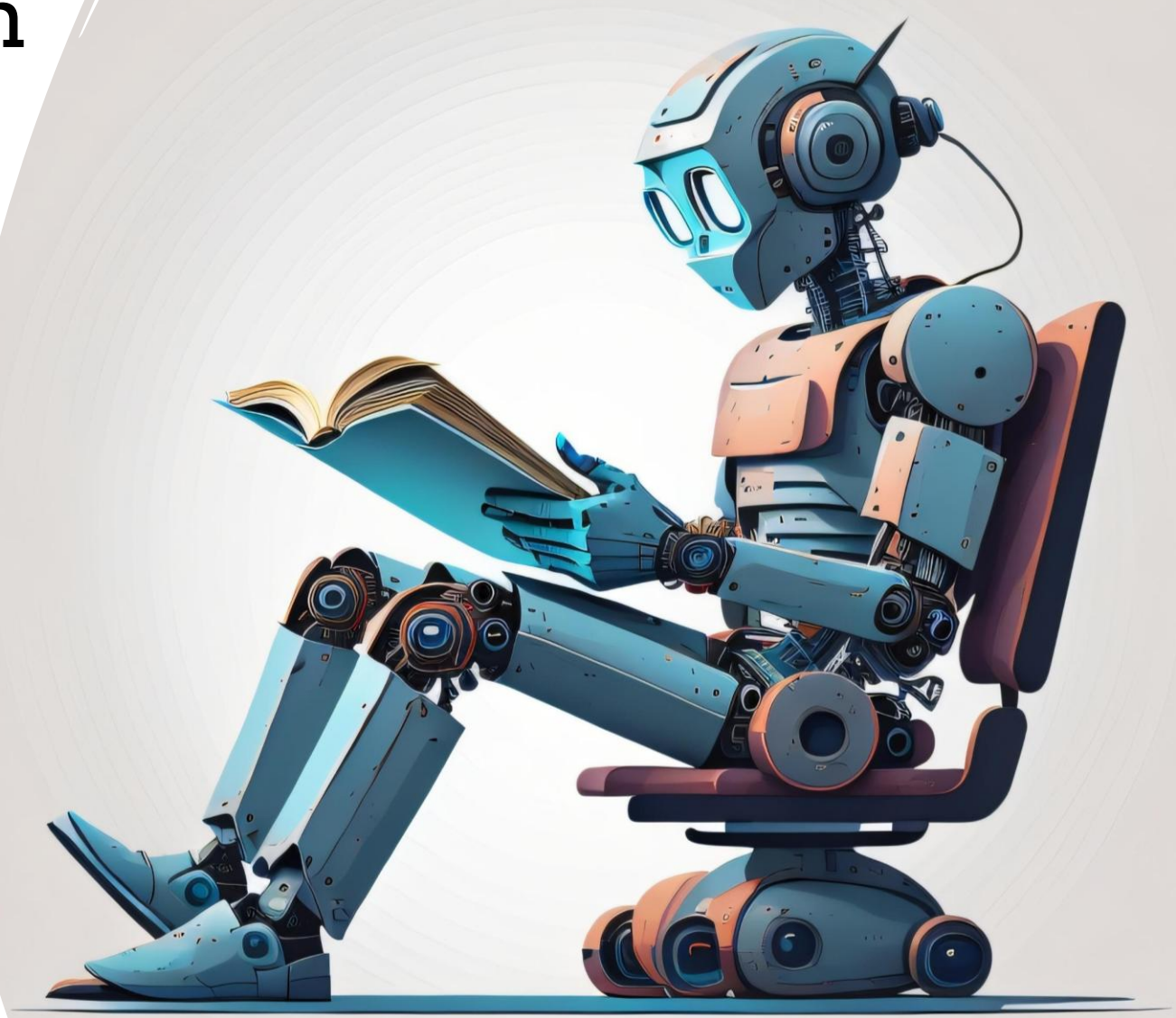
Book Recommendation with Self-Supervised Learning

TSM MachLeData – Spring 2026



Book Club:
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Jeanot Zubler, Sven Leutenegger

Motivation & Goal

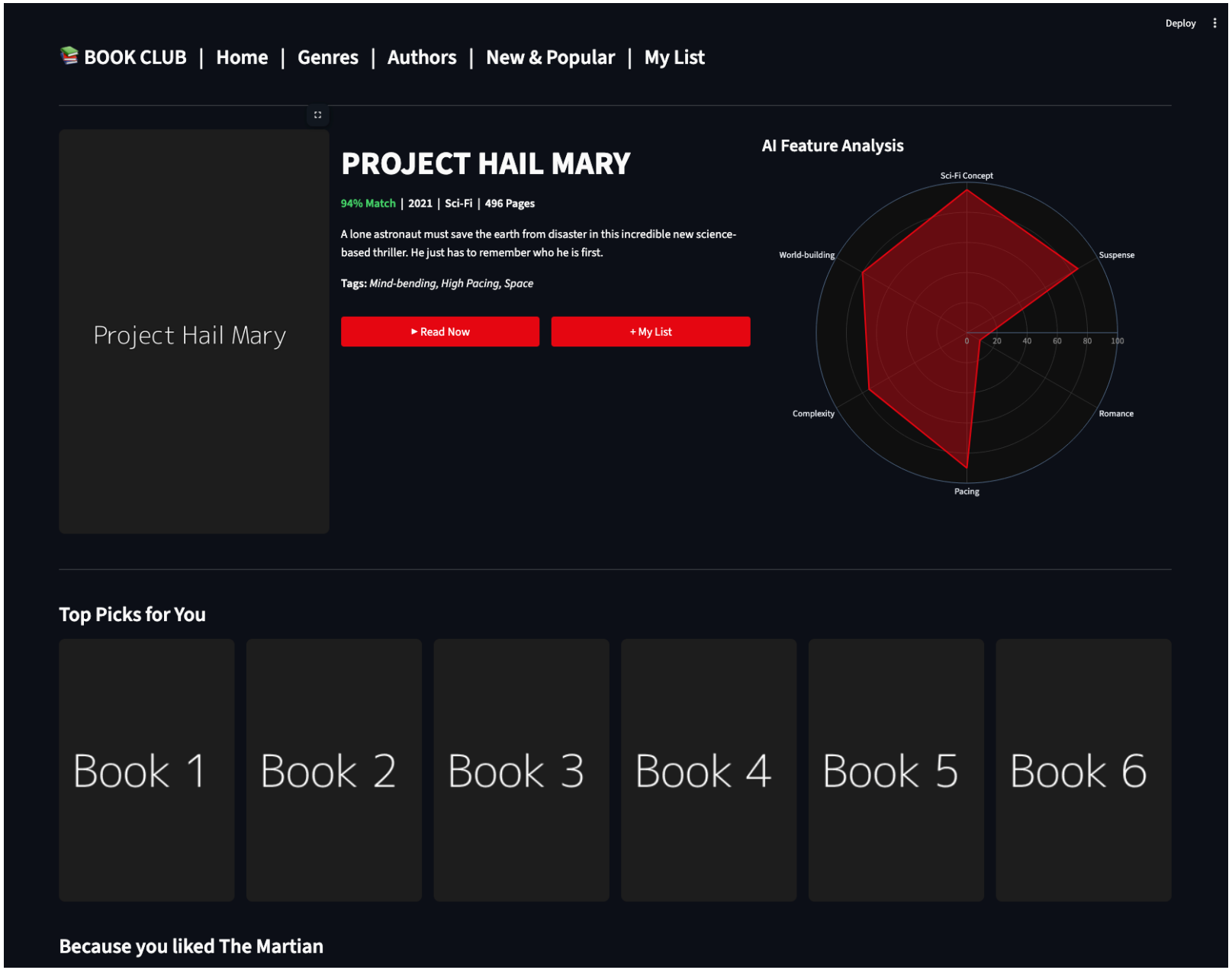
- 2-4 Million books published each year
- To find the gems in this sea of books is very hard
- **Our Goal:**
  Create a highly personalized, data-driven book recommendation system using self-supervised learning.

Achieving good Recommendations

- Use dataset from https://github.com/zygmuntz/goodbooks-10k
  - 10000 books most popular books on goodreads.com
  - ~6 Million ratings by 50000 users
  - Books marked as «to be read» by users

- Generate labels by excluding some of the ratings for each user

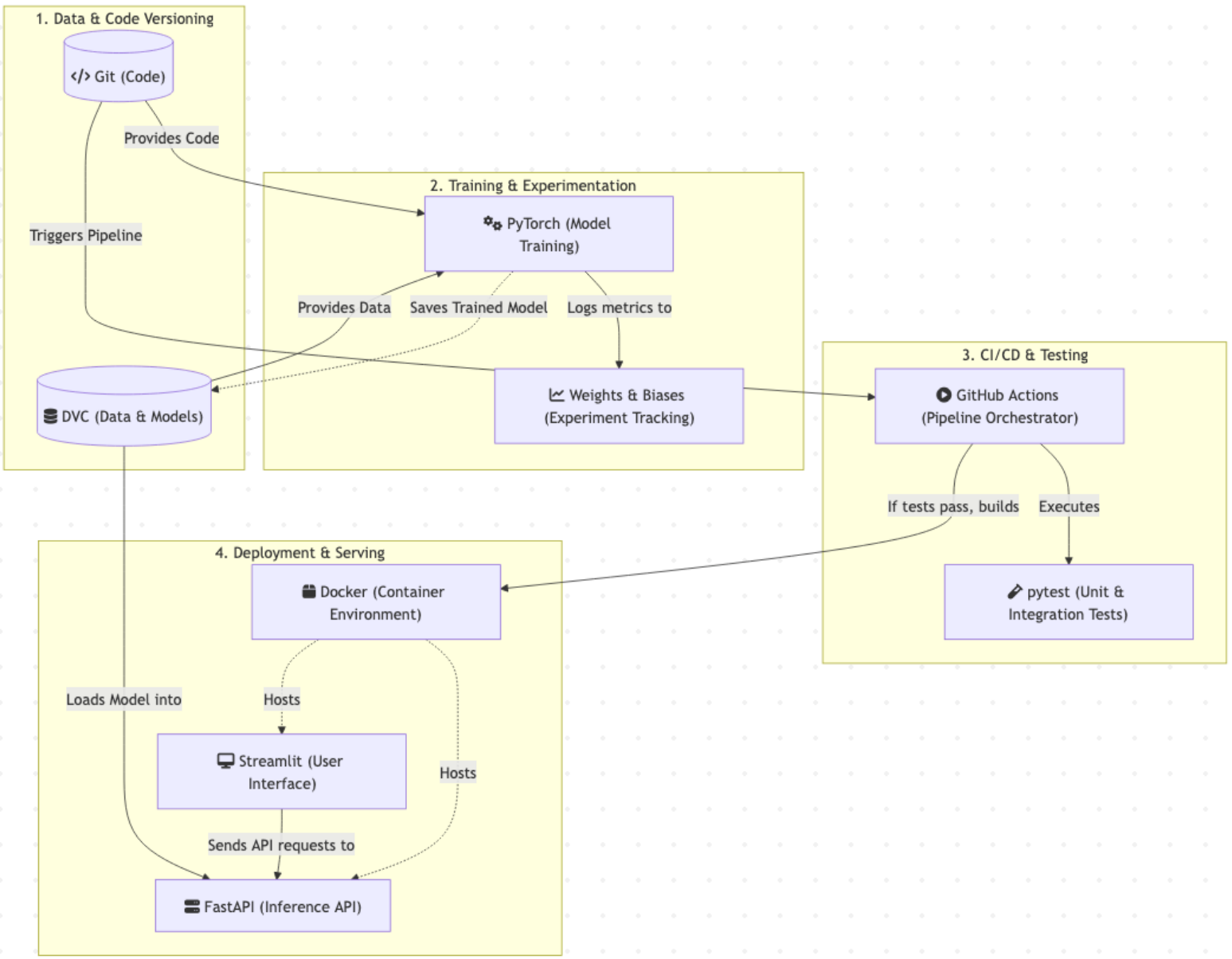
Mockup


Deploy
BOOK CLUB | Home | Genres | Authors | New & Popular | My List
Project Hail Mary
PROJECT HAIL MARY
94% Match | 2021 | Sci-Fi | 496 Pages
A lone astronaut must save the earth from disaster in this incredible new science-based thriller. He just has to remember who he is first.
Tags: Mind-bending, High Pacing, Space
▸ Read Now
+ My List
AI Feature Analysis
Sci-Fi Concept
Suspense
Romance
Pacing
Complexity
World-building
0
20
40
60
80
100
Top Picks for You
Book 1
Book 2
Book 3
Book 4
Book 5
Book 6
Because you liked The Martian

Technologies

Task	Tools
Versioning (data, model & code)	DVC git
Containerization	docker
Experiment tracking & tuning	W&B
Training	PyTorch
CI/CD & testing	GitHub Actions pytest
Deployment	FastAPI Streamlit

System Architecture



Versioning

- Code versioning in git
- Data and model versioning & validation in DVC
- DVC allows fast data and ML model access through built-in data caching
- Allows reproducibility
- Allows collaboration and parallel feature development

Training

W&B PyTorch

- Training and inference with PyTorch
- Track and visualize experiments with Weights & Biases
- Hyperparameter tuning with W&B

Testing

- Test different stages of ML system
  - Infrastructure tests for training
  - Training tests for data storage and training infrastructure
  - Functionality tests for prediction
  - …

- Automatically run tests in CI/CD pipeline

- Detect problems and regressions

Deployment

- Dockerized deployment for reproducibility and flexibility
- Use FastAPI to provide REST endpoints for model access
- Use Streamlit for a clean and simple UI